ALL QUESTIONS WILL BE SCALED UP TO 25 MARKS.

Total no. of Pages: ....2.........   Roll no. ....

Degree: ....B.Tech.......... Semester: ......5...Branch: MAC

MID-SEMESTER EXAMINATION, September, 2024

Course Title: ......Soft Computing................

Course Code: .... CMCSC16...........

Duration: 1:30 Hours   Max. Marks: 25

Q. No.	Question	Marks	CO
1a	Explain the motivation for developing a new domain of **soft computing**, as computing applications evolved over time.	**1.5**	**2**
1b	Explain the following paradoxes in Fuzzy Logic with example ? (i) A fuzzy set AND its compliment are not necessarily empty. (ii) A member of a fuzzy set can simultaneously be a member of another fuzzy set	**1.5**	**2**
2a	What is wrong with the fuzzy sets' depiction below: Fuzzy set A = {2, 5, 8, 10} Fuzzy set B = {(1.5, 0.8), (2.3, 0.9), (5.4, 0.1), (6, 1.2)}	**1.5**	**2**
2b	Given the following fuzzy sets, define the given logical operations: Outdated = {(1,0.8), (2,0.2),(3,0.9),(4,0.4)} Modern = {(1,0.6), (2,0.7),(3,0.5),(4,1)} (i) NOT outdated AND modern (ii) Outdated XOR Modern	**1.5**	**2**
3a	Think of any application that can be approached using the Mamdani model. Give a complete fuzzy description of: (i) its 2 input variables, each variable having 2 fuzzy classes (ii) 1 output variable having two fuzzy classes.	**1.5**	**3**
3b	For your application, describe a set of rules that govern the application.	**1.5**	**3**
4a	For the application you conceptualised in question 3, create a combination of input variable values and calculate the strength of each antecedent clause. Also calculate the strength of each rule.	**1.5**	**4**
4b	Perform the Rules aggregation step for each fuzzy class of the output, and find the clipped value of the output fuzzy classes.	**1.5**	**5**
5a	In the application of questions 3 and 4, evaluate the final crisp value of the output variable using any method of defuzzification	**1.5**	**5**
5b	Describe any chromosome encoding method for the play-list cum time table generation problem discussed in class. What are its objectives?	**1.5**	**3**